Author & Title	Year	Methodology	Key Contribution	Limitations
Dao & Dao – Lumename (2025)	2025	MFCC feature extraction and lightweight CNN deployed via TinyML on Arduino Nano BLE Sense. Personalized keyword detection with haptic alerts.	Privacy-preserving personalized sound detection with low latency and wearable deployment.	Detects only a single personalized keyword. No multi-class environmental sound recognition.
Zhong et al. – AHEAD-DS and YAMNet+	2025	Created AHEAD-DS dataset and enhanced YAMNet+ architecture for hearing-assistance devices with real-time Android deployment.	High accuracy environmental sound awareness with low inference latency.	Limited to predefined sound categories. No personalization.
Yeow et al. – Lightweight SELD	2025	Sound Event Localization and Detection model for wearable devices capable of identifying sound type and source direction.	Improves situational awareness through simultaneous classification and localization.	Does not support user-specific sound customization.
Liu et al. – YAMNet-Trans	2025	Transfer learning using pretrained YAMNet embeddings trained on ESC-50 and UrbanSound8K.	High environmental sound classification accuracy with limited labeled data.	No wearable implementation and no personalization.
ProtoSound	2022	Few-shot learning framework allowing users to teach custom sounds using only a few examples.	Enables scalable personalized sound recognition without full retraining.	Mobile-oriented system without wearable hardware deployment.
Environmental Sound Recognition on Embedded Devices	2025	Uses model compression, quantization and lightweight neural networks for embedded audio AI.	Provides deployment techniques for low-power sound recognition systems.	Focuses on deployment rather than personalized recognition.
ACDNet Edge ESC Pipeline	2021	Compact CNN architecture optimized for microcontrollers and resource-constrained edge devices.	High accuracy with very low computational cost.	Lacks personalization and adaptability.
Wearable Sound Detection for Hearing-Impaired Users	2018	Environmental sound recognition integrated with wearable notification hardware.	Early assistive sound awareness solution for hearing-impaired users.	Older machine learning techniques with lower accuracy.
Research Gap & Proposed Work (Sonic)	2026	Two-tier architecture using YAMNet transfer learning and personalized fingerprint matching.	Combines environmental sound recognition, personalization, wearable deployment and haptic alerts.	Addresses limitations of existing personalized and ESC systems.